Daniel Carrascoza

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Education

University of Southern California Expected Dec 2026
Master of Science in Electrical Engineering

University of California, San Diego June 2024
Bachelor of Science in Computer Engineering

Experience

Hardware Validation and Characterization *La Jolla, CA*
University of California, San Diego Sep 2023 – Jun 2024

- Designed and validated analog filters, amplifiers, and digital logic circuits using oscilloscopes, function generators, multimeters, and LTSpice-based simulation workflows.
- Performed frequency sweeps and waveform characterization to measure gain, phase shift, bandwidth, and distortion across varying input conditions.
- Debugged signal integrity and stability issues by correlating measured hardware behavior against simulated results, improving agreement between modeled and measured responses.
- Conducted DC operating point and noise analysis to optimize amplifier stability and measurement reliability across multiple circuit configurations.
- Captured and analyzed transient and steady-state waveforms across multiple analog and mixed-signal configurations using Tektronix TDS2002C oscilloscopes and AFG1022 signal generators.

Research Assistant *Los Angeles, CA*
University of Southern California Sep 2025 – May 2026

- Developed automated validation pipelines, regression checks, and failure-case analysis workflows in Python for machine learning experimentation.
- Analyzed preprocessing, model behavior, and data quality issues across large-scale medical imaging datasets.
- Documented experiment results, debugging findings, and evaluation metrics across iterative development cycles.

Projects

Network Inline Classification Engine | *NetFPGA, Verilog, Python, CUDA/PTX, Linux*

- Implemented an inline ML IDS SoC on NetFPGA with on-die packet classification and no host CPU; achieved 82.4% accuracy, 0.848 F1, and 0.864 attack recall on KDDTest+ (22,544 packets) at 7 ms/packet.
- Designed a custom 32-bit ISA with an ARM-style assembler and PTX compiler frontend, enabling CUDA kernel execution on a custom hardware datapath.
- Implemented a 5-stage quad-threaded CPU with a 4-lane BF16 SIMD tensor unit; added a cross-lane `BCAST` instruction achieving 3.28× speedup (2,402 → 732 cycles) measured using on-chip PCI counters.
- Validated 35 RTL test cases across IP block, pipeline, and system integration levels using waveform debugging and hardware characterization workflows.
- Measured 2.67 W power consumption at 90 nm and projected ∼100 mW ASIC power at 28 nm through scaling and performance analysis.

SqueezeNet Hardware Acceleration on Kria KV260 | *Xilinx Vitis HLS, FPGA, Python*

- Deployed a SqueezeNet FPGA accelerator using Vitis HLS with loop pipelining and unrolling; achieved 5× CPU speedup and ∼60% lower inference latency.
- Tuned HLS pragmas, memory interfaces, and dataflow optimizations to improve throughput and hardware utilization.
- Evaluated INT8 quantization tradeoffs across latency, throughput, and inference accuracy on embedded target hardware.
- Performed hardware/software validation and profiling across FPGA deployment workflows.

16-bit Multiply-and-Accumulate (MAC) Circuit | *Cadence Virtuoso, CMOS Logic, VLSI Design*

- Designed a 16-bit multiply-and-accumulate (MAC) circuit in Cadence Virtuoso using custom Booth encoders, 3:2 compressors, and 6:2 compressor datapaths for low-latency CMOS implementation.
- Completed schematic capture, symbol generation, DUT configuration, and full-custom physical layout across arithmetic datapath components.
- Executed DRC/LVS verification, parasitic extraction, and post-layout waveform validation across encoder and compressor blocks.
- Verified functional correctness through pre and post-extraction simulations under multiple vector test conditions using extracted timing behavior and waveform analysis.

Technical Skills

Programming: Python, MATLAB, C/C++, CUDA, Bash, Java, ARM/RISC-V Assembly
Lab & Validation: Oscilloscopes, Logic Analyzers, Function Generators, Multimeters, Waveforms, Signal Integrity
Digital Design: Verilog, SystemVerilog, RTL Verification, FPGA (NetFPGA, Kria KV260), Vitis HLS
VLSI & Analog: Cadence Virtuoso, CMOS Design, DRC/LVS, Parasitic Extraction, LTSpice
Systems: Linux, SSH, Git, GDB, perf